



BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)

FINAL EXAMINATION JUNE 2025

Course: EC3122 (Web Development)
Lecturer: Muhammad Hamizan Johari

Time: 2.00 pm – 5.00 pm (3 hours)
Date: 21 November 2025

Instructions:

Answer **ALL** questions.

The maximum number of marks is 100.

This question paper consists of **4** printed pages
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Briefly explain the following multimedia elements.

- i. Animation (2 MARKS)
- ii. Audio (2 MARKS)
- iii. Images (2 MARKS)
- iv. Text (2 MARKS)
- v. Video (2 MARKS)

b. Briefly describe the following components related to web development.

- i. Hyperlink (2 MARKS)
- ii. Client (2 MARKS)
- iii. Server (2 MARKS)

c. Briefly discuss at least **THREE (3)** main website designs used in web development. (9 MARKS)

2. a. Briefly explain each of the following HTML semantic elements.

- i. <head> tag (2 MARKS)
- ii. <footer> tag (2 MARKS)

Order Num.	Item	Amount
001	Smartphone	35
002	Laptop	22
004	Tablet	1

Table 1: Ordering Table

b. Write a HTML code to create the table given above. (15 MARKS)

Menu Num.	Menu	Description
11	Cappuccino	Hot espresso with steamed milk foam
32	Chicken Sandwich	Grilled chicken with lettuce and mayo on toasted bread
74	Blueberry Muffin	Freshly baked muffin with blueberries

Table 2: Café Menu

- c. Write a HTML code that describe the menu above in *description list* format. **(6 MARKS)**
3. a. Explain **THREE (3)** data types of variables for JavaScript. Give at least **ONE (1)** example of the data types. **(9 MARKS)**
- b. Rewrite the following *switch case statement* into an *if...else statement*. **(6 MARKS)**

```

switch (true) {
  case (marks >= 90):
    grade = "A+";
    break;
  case (marks >= 80):
    grade = "A";
    break;
  case (marks >= 70):
    grade = "B";
    break;
  case (marks >= 60):
    grade = "C";
    break;
  case (marks >= 50):
    grade = "D";
    break;
  default:
    grade = "Fail";
}

```

- c. Write the output of the *nested for loop* statement below.

(6 MARKS)

```
const places = ["Kathmandu", "Pokhara", "Lumbini", "Chitwan"];

// Nested for loop to display pairs of places
for (let i = 0; i < places.length; i++) {
  for (let j = i + 1; j < places.length; j++) {
    console.log(`${places[i]} and ${places[j]}`);
  }
}
```

- d. Write all the values of *i* of the *do...while* statement.

(4 MARKS)

```
let i = 0;

do {
  console.log ("Current value of i:", i);
  i++;
} while (i < 6);
```

4. a. Briefly explain the following types of CSS selectors.

i. Class selector.

(2 MARKS)

ii. Element selector.

(2 MARKS)

iii. ID selector.

(2 MARKS)

- b. A website requirement requires a programmer to code a specific styling: *blue background colour*, *red text* and *green heading*. Write a CSS code for each of the given way of inserting CSS into a HTML file.

i. Inline

(5 MARKS)

ii. Internal

(5 MARKS)

iii. External

(5 MARKS)

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c. Write the animation output from the given CSS animation code.

(4 MARKS)

- ```
<style>
/* Define the animation */
@keyframes moveBox {
 from { transform: translateX(0px); }
 to { transform: translateX(200px); }
}

/* Apply the animation to an element */
.box {
 width: 100px;
 height: 100px;
 background-color: orange;
 animation: moveBox 2s infinite alternate;
}
</style>
```

**-END OF QUESTION PAPER-**